

WATER CHEMICAL VALIDATION REPORT FOR ICP SAMPLES

This report summarises the validation results of water sample data. It compares the original non-repeated sample validation results against a scenario where the validation results are replaced by their replicate (REP) counterparts when a REP exists. The comparison keeps the same number of samples in both scenarios. Separate charts are also provided for all samples and replicate-only samples.

Notes on the interpretation of quality criteria for ion balance and conductivity

- a) Quality criteria for conductivity must always be satisfied (OK) for each type of sample: wet only, bulk open field, throughfall, stemflow, soil water and runoff.
- b) Quality criteria for ion balance should be satisfied (OK) for open field samples (wet only and bulk) and runoff with low organic carbon concentrations. The ion balance criterion is not considered for throughfall and stemflow samples because of the presence of organic anions.
- c) Quality criteria for ON must always be satisfied (OK) for each type of sample: open field, throughfall, stemflow, soil water and runoff.
- d) Quality criteria for Na/Cl ratio (marine ratio = 0.86, accepted range 0.5-1.5) should be satisfied for each type of sample, excluding soil water and runoff samples.

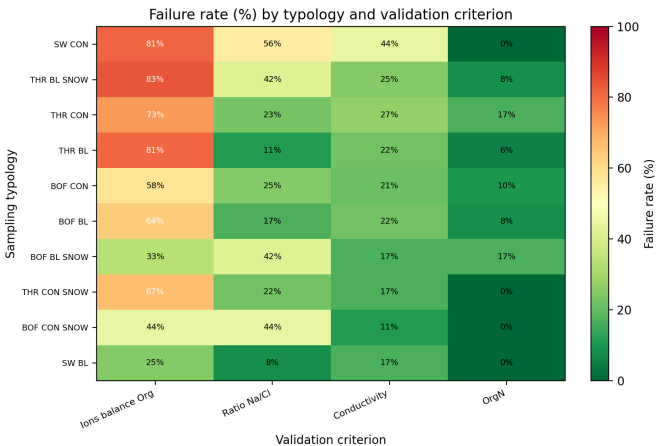
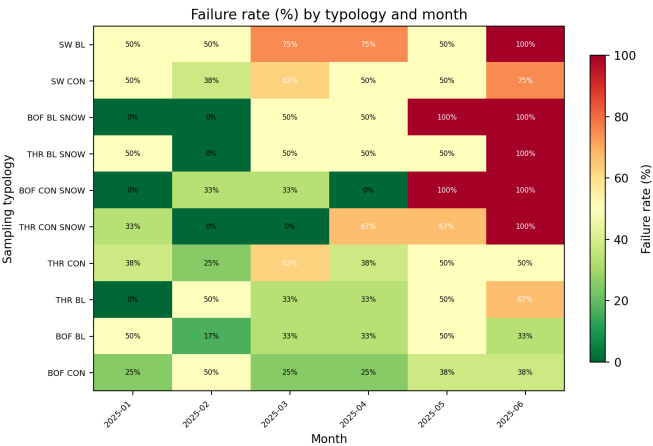
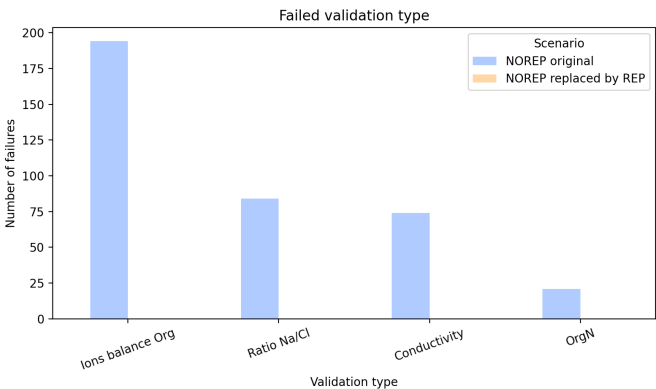
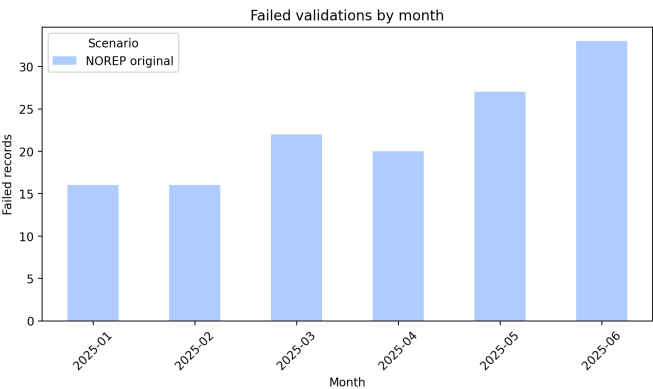
1.2 Water subprogrammes

Subprogram	Code	Description
Precipitation Chemistry	PC	Bulk deposition in open areas - quantifies wet deposition inputs
Throughfall	TF	Water passing through the forest canopy - reflects canopy interactions
Stemflow	SF	Water flowing down tree stems - localised flux insights
Soil Water	SW	Soil solution chemistry - acidification and nitrogen dynamics
Runoff Water	RW	Catchment outflows - estimates element export via hydrology

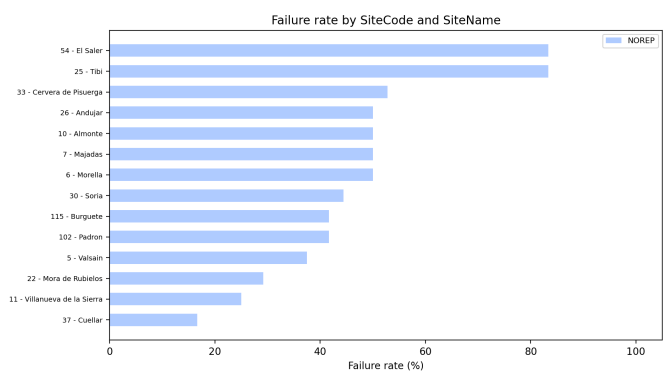
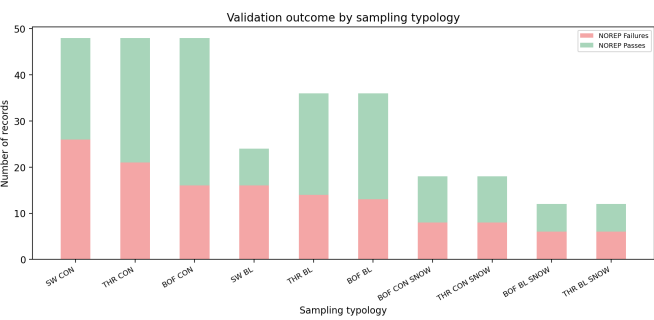
Key Indicators - NOREP Original

Metric	Value
Total records	300
Sites	14
Validated periods	6
Failed records	134
Passed records	166
Failure rate	44.7%
Unknown VAL	0

Graphical Summary - Overview and Heatmaps



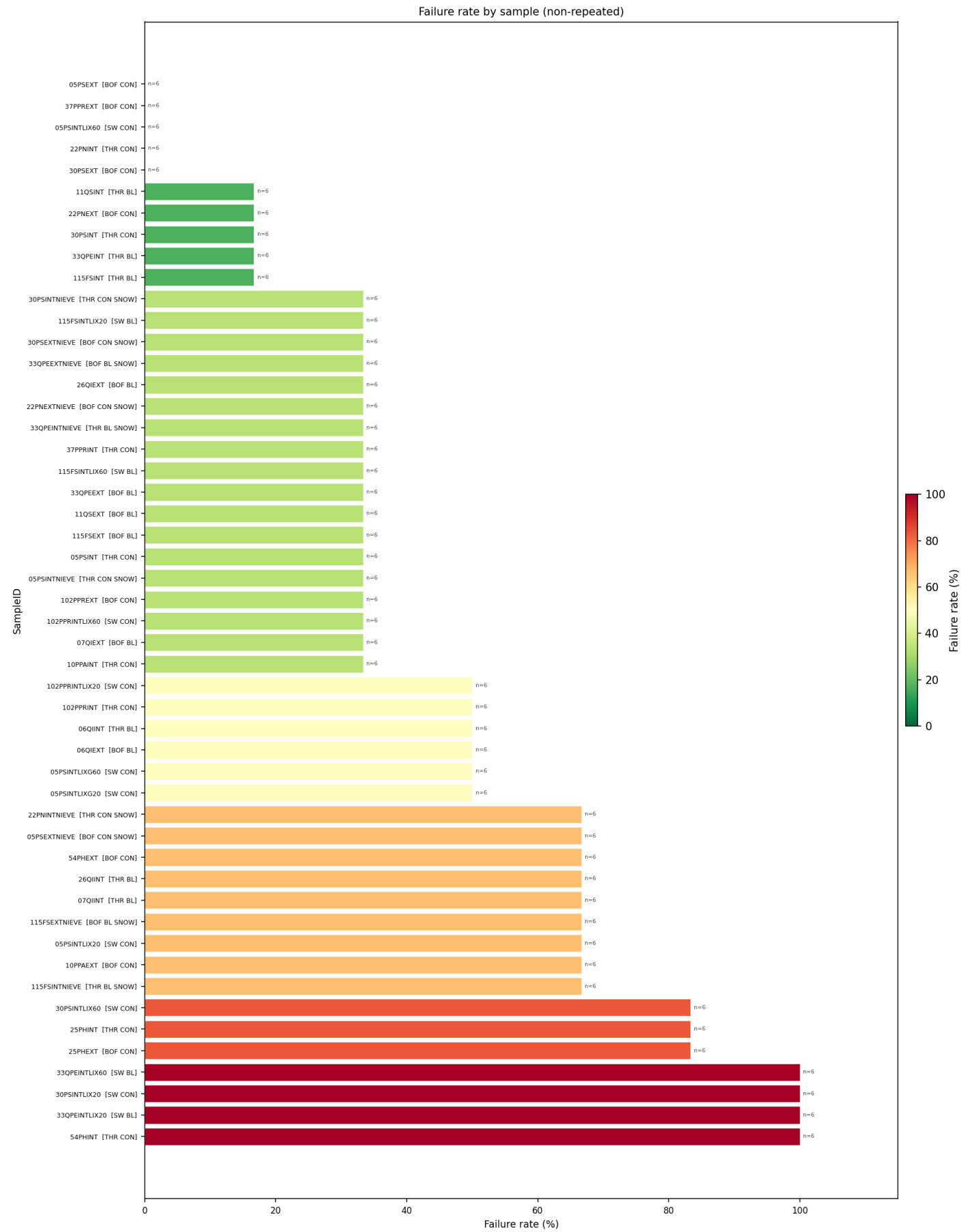
Graphical Summary - Typology, Site and REP Effects



REP improvement by typology (no REP data)

No data available

Validation Outcome by Non-Repeated Samples



Validation Table by Sample

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
05PSINT	1	THR CON	OK	OK	OK	OK	No Rep	SI
05PSINTNIEVE	1	THR CON SNOW	NO	OK	OK	OK	No Rep	SI
05PSINTLIX20	1	SW CON	NO	NO	OK	NO	No Rep	NO
05PSINTLIX60	1	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG20	1	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG60	1	SW CON	NO	NO	OK	OK	No Rep	NO
05PSEXT	1	BOF CON	OK	OK	OK	OK	No Rep	SI
05PSEXTNIEVE	1	BOF CON SNOW	OK	OK	OK	OK	No Rep	SI
06QIINT	1	THR BL	NO	OK	OK	OK	No Rep	SI
06QIEXT	1	BOF BL	NO	NO	OK	NO	No Rep	NO
07QIINT	1	THR BL	NO	OK	OK	OK	No Rep	SI
07QIEXT	1	BOF BL	NO	OK	OK	OK	No Rep	SI
10PPAINT	1	THR CON	NO	OK	OK	OK	No Rep	SI
10PPAEXT	1	BOF CON	OK	OK	OK	OK	No Rep	SI
11QSINT	1	THR BL	NO	OK	OK	OK	No Rep	SI
11QSEXT	1	BOF BL	OK	OK	OK	OK	No Rep	SI
22PNINT	1	THR CON	NO	OK	OK	OK	No Rep	SI
22PNINTNIEVE	1	THR CON SNOW	NO	NO	OK	OK	No Rep	NO
22PNEXT	1	BOF CON	NO	OK	OK	OK	No Rep	SI
22PNEXTNIEVE	1	BOF CON SNOW	NO	OK	OK	NO	No Rep	SI
25PHINT	1	THR CON	NO	OK	NO TN	OK	No Rep	NO
25PHEXT	1	BOF CON	NO	NO	OK	OK	No Rep	NO
26QIINT	1	THR BL	NO	OK	OK	OK	No Rep	SI
26QIEXT	1	BOF BL	NO	OK	OK	NO	No Rep	SI
30PSINT	1	THR CON	NO	NO	OK	OK	No Rep	NO
30PSINTNIEVE	1	THR CON SNOW	NO	OK	OK	NO	No Rep	SI
30PSINTLIX20	1	SW CON	NO	NO	NA	NO	No Rep	NO
30PSINTLIX60	1	SW CON	NO	NO	NA	NO	No Rep	NO
30PSEXT	1	BOF CON	OK	OK	OK	OK	No Rep	SI
30PSEXTNIEVE	1	BOF CON SNOW	OK	OK	OK	OK	No Rep	SI

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
33QPEINT	1	THR BL	OK	OK	OK	NO	No Rep	SI
33QPEINTNIEVE	1	THR BL SNOW	NO	OK	OK	NO	No Rep	SI
33QPEINTLIX20	1	SW BL	NA	NO	NA	NA	No Rep	NO
33QPEINTLIX60	1	SW BL	NA	NO	NA	NA	No Rep	NO
33QPEEXT	1	BOF BL	OK	OK	NO TN	OK	No Rep	NO
33QPEEXTNIEVE	1	BOF BL SNOW	NO	OK	OK	NO	No Rep	SI
37PPRINT	1	THR CON	NO	OK	OK	OK	No Rep	SI
37PPREXT	1	BOF CON	NO	OK	OK	OK	No Rep	SI
54PHINT	1	THR CON	NO	NO	OK	NO	No Rep	NO
54PHEXT	1	BOF CON	NO	OK	OK	NO	No Rep	SI
102PPRINT	1	THR CON	OK	OK	OK	OK	No Rep	SI
102PPRINTLIX20	1	SW CON	OK	OK	OK	OK	No Rep	SI
102PPRINTLIX60	1	SW CON	OK	OK	OK	NO	No Rep	SI
102PPREXT	1	BOF CON	NO	NO	OK	NO	No Rep	NO
115FSINT	1	THR BL	OK	OK	OK	OK	No Rep	SI
115FSINTNIEVE	1	THR BL SNOW	NO	NO	OK	NO	No Rep	NO
115FSINTLIX20	1	SW BL	OK	OK	OK	NO	No Rep	SI
115FSINTLIX60	1	SW BL	OK	OK	OK	OK	No Rep	SI
115FSEXT	1	BOF BL	NO	NO	OK	OK	No Rep	NO
115FSEXTNIEVE	1	BOF BL SNOW	OK	OK	OK	OK	No Rep	SI
05PSINT	2	THR CON	OK	OK	OK	OK	No Rep	SI
05PSINTNIEVE	2	THR CON SNOW	NO	OK	OK	OK	No Rep	SI
05PSINTLIX20	2	SW CON	NO	NO	OK	OK	No Rep	NO
05PSINTLIX60	2	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG20	2	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG60	2	SW CON	OK	OK	OK	NO	No Rep	SI
05PSEXT	2	BOF CON	OK	OK	OK	OK	No Rep	SI
05PSEXTNIEVE	2	BOF CON SNOW	NO	NO	OK	NO	No Rep	NO
06QIINT	2	THR BL	NA	NA	NA	NA	No Rep	NO
06QIEXT	2	BOF BL	NO	NO	NA	OK	No Rep	NO

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
07QIINT	2	THR BL	NO	OK	NO TN	OK	No Rep	NO
07QIEXT	2	BOF BL	OK	OK	OK	OK	No Rep	SI
10PPAINT	2	THR CON	NO	OK	OK	OK	No Rep	SI
10PPAEXT	2	BOF CON	NO	NO	OK	OK	No Rep	NO
11QSINT	2	THR BL	NO	OK	OK	OK	No Rep	SI
11QSEXT	2	BOF BL	OK	OK	OK	OK	No Rep	SI
22PNINT	2	THR CON	NO	OK	OK	OK	No Rep	SI
22PNINTNIEVE	2	THR CON SNOW	NO	OK	OK	OK	No Rep	SI
22PNEXT	2	BOF CON	NO	NO	OK	OK	No Rep	NO
22PNEXTNIEVE	2	BOF CON SNOW	NO	OK	OK	NO	No Rep	SI
25PHINT	2	THR CON	NA	NA	NA	NA	No Rep	NO
25PHEXT	2	BOF CON	NA	NA	NA	NA	No Rep	NO
26QIINT	2	THR BL	NO	NO	OK	OK	No Rep	NO
26QIEXT	2	BOF BL	OK	OK	OK	OK	No Rep	SI
30PSINT	2	THR CON	NO	OK	OK	OK	No Rep	SI
30PSINTNIEVE	2	THR CON SNOW	NO	OK	OK	OK	No Rep	SI
30PSINTLIX20	2	SW CON	NO	NO	OK	NO	No Rep	NO
30PSINTLIX60	2	SW CON	NO	NO	OK	OK	No Rep	NO
30PSEXT	2	BOF CON	OK	OK	OK	OK	No Rep	SI
30PSEXTNIEVE	2	BOF CON SNOW	OK	OK	OK	NO	No Rep	SI
33QPEINT	2	THR BL	NO	OK	OK	OK	No Rep	SI
33QPEINTNIEVE	2	THR BL SNOW	NO	OK	OK	OK	No Rep	SI
33QPEINTLIX20	2	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEINTLIX60	2	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEEXT	2	BOF BL	NO	OK	OK	OK	No Rep	SI
33QPEEXTNIEVE	2	BOF BL SNOW	NO	OK	OK	OK	No Rep	SI
37PPRINT	2	THR CON	NO	OK	OK	OK	No Rep	SI
37PPREXT	2	BOF CON	OK	OK	OK	OK	No Rep	SI
54PHINT	2	THR CON	NO	NO	NO TN	OK	No Rep	NO
54PHEXT	2	BOF CON	OK	OK	NO TN	OK	No Rep	NO

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
102PPRINT	2	THR CON	OK	OK	OK	OK	No Rep	SI
102PPRINTLIX20	2	SW CON	NO	OK	OK	OK	No Rep	SI
102PPRINTLIX60	2	SW CON	OK	OK	OK	OK	No Rep	SI
102PPREXT	2	BOF CON	OK	OK	OK	OK	No Rep	SI
115FSINT	2	THR BL	NO	OK	OK	OK	No Rep	SI
115FSINTNIEVE	2	THR BL SNOW	NO	OK	OK	OK	No Rep	SI
115FSINTLIX20	2	SW BL	NO	OK	OK	OK	No Rep	SI
115FSINTLIX60	2	SW BL	OK	OK	OK	OK	No Rep	SI
115FSEXT	2	BOF BL	NO	OK	OK	OK	No Rep	SI
115FSEXTNIEVE	2	BOF BL SNOW	OK	OK	OK	OK	No Rep	SI
05PSINT	3	THR CON	NO	NO	OK	NO	No Rep	NO
05PSINTNIEVE	3	THR CON SNOW	NO	OK	OK	OK	No Rep	SI
05PSINTLIX20	3	SW CON	NO	NO	OK	NO	No Rep	NO
05PSINTLIX60	3	SW CON	NO	OK	OK	OK	No Rep	SI
05PSINTLIXG20	3	SW CON	NO	NO	OK	OK	No Rep	NO
05PSINTLIXG60	3	SW CON	NO	NO	OK	NO	No Rep	NO
05PSEXT	3	BOF CON	NO	OK	OK	OK	No Rep	SI
05PSEXTNIEVE	3	BOF CON SNOW	NO	NO	OK	OK	No Rep	NO
06QIINT	3	THR BL	NO	OK	OK	OK	No Rep	SI
06QIEXT	3	BOF BL	OK	OK	OK	OK	No Rep	SI
07QIINT	3	THR BL	NO	OK	OK	OK	No Rep	SI
07QIEXT	3	BOF BL	OK	OK	OK	OK	No Rep	SI
10PPAINT	3	THR CON	NO	OK	OK	OK	No Rep	SI
10PPAEXT	3	BOF CON	OK	OK	OK	OK	No Rep	SI
11QSINT	3	THR BL	NO	OK	OK	OK	No Rep	SI
11QSEXT	3	BOF BL	NO	OK	OK	OK	No Rep	SI
22PNINT	3	THR CON	NO	OK	OK	OK	No Rep	SI
22PNINTNIEVE	3	THR CON SNOW	OK	OK	OK	NO	No Rep	SI
22PNEXT	3	BOF CON	OK	OK	OK	OK	No Rep	SI
22PNEXTNIEVE	3	BOF CON SNOW	OK	OK	OK	NO	No Rep	SI

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
25PHINT	3	THR CON	NO	NO	OK	NO	No Rep	NO
25PHEXT	3	BOF CON	NO	NO	OK	NO	No Rep	NO
26QIINT	3	THR BL	NO	NO	NA	NA	No Rep	NO
26QIEXT	3	BOF BL	OK	OK	OK	OK	No Rep	SI
30PSINT	3	THR CON	NO	OK	OK	NO	No Rep	SI
30PSINTNIEVE	3	THR CON SNOW	NO	OK	OK	NO	No Rep	SI
30PSINTLIX20	3	SW CON	NO	NO	OK	OK	No Rep	NO
30PSINTLIX60	3	SW CON	NO	OK	OK	OK	No Rep	SI
30PSEXT	3	BOF CON	OK	OK	OK	NO	No Rep	SI
30PSEXTNIEVE	3	BOF CON SNOW	NO	OK	OK	NO	No Rep	SI
33QPEINT	3	THR BL	OK	OK	OK	NO	No Rep	SI
33QPEINTNIEVE	3	THR BL SNOW	NO	OK	OK	NO	No Rep	SI
33QPEINTLIX20	3	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEINTLIX60	3	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEEXT	3	BOF BL	NO	OK	NA	NO	No Rep	NO
33QPEEXTNIEVE	3	BOF BL SNOW	OK	OK	OK	NO	No Rep	SI
37PPRINT	3	THR CON	OK	NO	OK	NO	No Rep	NO
37PPREXT	3	BOF CON	NO	OK	OK	OK	No Rep	SI
54PHINT	3	THR CON	NO	NO	NO TN	NO	No Rep	NO
54PHEXT	3	BOF CON	NO	NO	NO TN	NO	No Rep	NO
102PPRINT	3	THR CON	OK	NO	OK	OK	No Rep	NO
102PPRINTLIX20	3	SW CON	NO	NO	OK	OK	No Rep	NO
102PPRINTLIX60	3	SW CON	NO	OK	OK	OK	No Rep	SI
102PPREXT	3	BOF CON	OK	OK	OK	OK	No Rep	SI
115FSINT	3	THR BL	NO	NO	OK	NO	No Rep	NO
115FSINTNIEVE	3	THR BL SNOW	NO	NO	OK	OK	No Rep	NO
115FSINTLIX20	3	SW BL	NO	OK	OK	OK	No Rep	SI
115FSINTLIX60	3	SW BL	NO	NO	OK	NO	No Rep	NO
115FSEXT	3	BOF BL	NO	NO	OK	NO	No Rep	NO
115FSEXTNIEVE	3	BOF BL SNOW	NO	NO	OK	NO	No Rep	NO

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
05PSINT	4	THR CON	OK	OK	OK	OK	No Rep	SI
05PSINTNIEVE	4	THR CON SNOW	NO	NO	OK	OK	No Rep	NO
05PSINTLIX20	4	SW CON	NO	NO	OK	NO	No Rep	NO
05PSINTLIX60	4	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG20	4	SW CON	NO	NO	OK	NO	No Rep	NO
05PSINTLIXG60	4	SW CON	NO	OK	OK	NO	No Rep	SI
05PSEXT	4	BOF CON	NO	OK	OK	NO	No Rep	SI
05PSEXTNIEVE	4	BOF CON SNOW	NO	OK	OK	NO	No Rep	SI
06QIINT	4	THR BL	NO	NO	OK	OK	No Rep	NO
06QIEXT	4	BOF BL	NO	OK	OK	OK	No Rep	SI
07QIINT	4	THR BL	NO	NO	OK	OK	No Rep	NO
07QIEXT	4	BOF BL	NO	NO	NO TN	OK	No Rep	NO
10PPAINT	4	THR CON	NO	NO	OK	NO	No Rep	NO
10PPAEXT	4	BOF CON	NO	NO	OK	NO	No Rep	NO
11QSINT	4	THR BL	NO	OK	OK	NA	No Rep	SI
11QSEXT	4	BOF BL	NO	OK	OK	NO	No Rep	SI
22PNINT	4	THR CON	NO	OK	OK	OK	No Rep	SI
22PNINTNIEVE	4	THR CON SNOW	NO	NO	OK	OK	No Rep	NO
22PNEXT	4	BOF CON	NO	OK	OK	OK	No Rep	SI
22PNEXTNIEVE	4	BOF CON SNOW	NO	OK	OK	OK	No Rep	SI
25PHINT	4	THR CON	NO	OK	OK	OK	No Rep	SI
25PHEXT	4	BOF CON	NO	NO	OK	NO	No Rep	NO
26QIINT	4	THR BL	NO	OK	OK	OK	No Rep	SI
26QIEXT	4	BOF BL	OK	NO	OK	OK	No Rep	NO
30PSINT	4	THR CON	OK	OK	OK	NO	No Rep	SI
30PSINTNIEVE	4	THR CON SNOW	NO	OK	OK	OK	No Rep	SI
30PSINTLIX20	4	SW CON	NO	NO	OK	NO	No Rep	NO
30PSINTLIX60	4	SW CON	NO	NO	OK	NO	No Rep	NO
30PSEXT	4	BOF CON	OK	OK	OK	OK	No Rep	SI
30PSEXTNIEVE	4	BOF CON SNOW	NO	OK	OK	NO	No Rep	SI

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
33QPEINT	4	THR BL	OK	OK	OK	OK	No Rep	SI
33QPEINTNIEVE	4	THR BL SNOW	NO	OK	OK	NO	No Rep	SI
33QPEINTLIX20	4	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEINTLIX60	4	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEEXT	4	BOF BL	NO	OK	OK	NA	No Rep	SI
33QPEEXTNIEVE	4	BOF BL SNOW	OK	OK	OK	NO	No Rep	SI
37PPRINT	4	THR CON	NO	OK	OK	OK	No Rep	SI
37PPREXT	4	BOF CON	NO	OK	OK	OK	No Rep	SI
54PHINT	4	THR CON	OK	NO	OK	OK	No Rep	NO
54PHEXT	4	BOF CON	NO	OK	OK	OK	No Rep	SI
102PPRINT	4	THR CON	NO	NO	OK	OK	No Rep	NO
102PPRINTLIX20	4	SW CON	NO	OK	OK	OK	No Rep	SI
102PPRINTLIX60	4	SW CON	NO	OK	OK	OK	No Rep	SI
102PPREXT	4	BOF CON	NO	OK	OK	OK	No Rep	SI
115FSINT	4	THR BL	NO	OK	OK	OK	No Rep	SI
115FSINTNIEVE	4	THR BL SNOW	NO	NO	OK	OK	No Rep	NO
115FSINTLIX20	4	SW BL	NO	NO	OK	OK	No Rep	NO
115FSINTLIX60	4	SW BL	NO	OK	OK	OK	No Rep	SI
115FSEXT	4	BOF BL	NO	OK	OK	OK	No Rep	SI
115FSEXTNIEVE	4	BOF BL SNOW	NO	OK	NO TN	OK	No Rep	NO
05PSINT	5	THR CON	NO	OK	OK	NO	No Rep	SI
05PSINTNIEVE	5	THR CON SNOW	NO	OK	OK	NO	No Rep	SI
05PSINTLIX20	5	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIX60	5	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG20	5	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG60	5	SW CON	NO	OK	OK	NO	No Rep	SI
05PSEXT	5	BOF CON	NO	OK	OK	NO	No Rep	SI
05PSEXTNIEVE	5	BOF CON SNOW	NA	NA	NA	NA	No Rep	NO
06QIINT	5	THR BL	NO	OK	OK	OK	No Rep	SI
06QIEXT	5	BOF BL	NO	OK	NO TN	OK	No Rep	NO

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
07QIINT	5	THR BL	NO	NO	OK	OK	No Rep	NO
07QIEXT	5	BOF BL	NO	OK	OK	OK	No Rep	SI
10PPAINT	5	THR CON	NO	OK	OK	OK	No Rep	SI
10PPAEXT	5	BOF CON	OK	NO	NO TN	OK	No Rep	NO
11QSINT	5	THR BL	NO	OK	OK	OK	No Rep	SI
11QSEXT	5	BOF BL	NO	NO	OK	OK	No Rep	NO
22PNINT	5	THR CON	NO	OK	OK	NO	No Rep	SI
22PNINTNIEVE	5	THR CON SNOW	NA	NA	NA	NA	No Rep	NO
22PNEXT	5	BOF CON	NO	OK	OK	NO	No Rep	SI
22PNEXTNIEVE	5	BOF CON SNOW	NA	NA	NA	NA	No Rep	NO
25PHINT	5	THR CON	OK	NO	NO TN	OK	No Rep	NO
25PHEXT	5	BOF CON	NO	OK	OK	OK	No Rep	SI
26QIINT	5	THR BL	NO	OK	NO TN	OK	No Rep	NO
26QIEXT	5	BOF BL	NO	NO	OK	OK	No Rep	NO
30PSINT	5	THR CON	NO	OK	OK	OK	No Rep	SI
30PSINTNIEVE	5	THR CON SNOW	NA	NA	NA	NA	No Rep	NO
30PSINTLIX20	5	SW CON	NO	NO	OK	OK	No Rep	NO
30PSINTLIX60	5	SW CON	NO	NO	OK	NO	No Rep	NO
30PSEXT	5	BOF CON	NO	OK	OK	OK	No Rep	SI
30PSEXTNIEVE	5	BOF CON SNOW	NA	NA	NA	NA	No Rep	NO
33QPEINT	5	THR BL	NO	OK	NA	NO	No Rep	NO
33QPEINTNIEVE	5	THR BL SNOW	NO	OK	NO TN	OK	No Rep	NO
33QPEINTLIX20	5	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEINTLIX60	5	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEEXT	5	BOF BL	NO	OK	OK	NO	No Rep	SI
33QPEEXTNIEVE	5	BOF BL SNOW	OK	OK	NO TN	NO	No Rep	NO
37PPRINT	5	THR CON	NO	OK	NO TN	OK	No Rep	NO
37PPREXT	5	BOF CON	OK	OK	OK	OK	No Rep	SI
54PHINT	5	THR CON	NO	OK	NO TN	OK	No Rep	NO
54PHEXT	5	BOF CON	NO	NO	OK	OK	No Rep	NO

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
102PPRINT	5	THR CON	NO	OK	NO TN	OK	No Rep	NO
102PPRINTLIX20	5	SW CON	NA	NA	NA	NA	No Rep	NO
102PPRINTLIX60	5	SW CON	NA	NA	NA	NA	No Rep	NO
102PPREXT	5	BOF CON	OK	OK	NO TN	OK	No Rep	NO
115FSINT	5	THR BL	NO	OK	OK	OK	No Rep	SI
115FSINTNIEVE	5	THR BL SNOW	NO	OK	OK	NO	No Rep	SI
115FSINTLIX20	5	SW BL	NO	OK	OK	OK	No Rep	SI
115FSINTLIX60	5	SW BL	OK	OK	OK	OK	No Rep	SI
115FSEXT	5	BOF BL	OK	OK	OK	OK	No Rep	SI
115FSEXTNIEVE	5	BOF BL SNOW	OK	NO	OK	OK	No Rep	NO
05PSINT	6	THR CON	NO	NO	OK	NO	No Rep	NO
05PSINTNIEVE	6	THR CON SNOW	NA	NA	NA	NA	No Rep	NO
05PSINTLIX20	6	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIX60	6	SW CON	NO	OK	OK	NO	No Rep	SI
05PSINTLIXG20	6	SW CON	NO	NO	OK	NO	No Rep	NO
05PSINTLIXG60	6	SW CON	NA	NA	NA	NA	No Rep	NO
05PSEXT	6	BOF CON	NO	OK	OK	OK	No Rep	SI
05PSEXTNIEVE	6	BOF CON SNOW	NA	NA	NA	NA	No Rep	NO
06QIINT	6	THR BL	NO	NO	OK	OK	No Rep	NO
06QIEXT	6	BOF BL	NO	OK	OK	OK	No Rep	SI
07QIINT	6	THR BL	NA	NA	NA	NA	No Rep	NO
07QIEXT	6	BOF BL	NA	NA	NA	NA	No Rep	NO
10PPAINT	6	THR CON	NA	NA	NA	NA	No Rep	NO
10PPAEXT	6	BOF CON	NA	NA	NA	NA	No Rep	NO
11QSINT	6	THR BL	NA	NA	NA	NA	No Rep	NO
11QSEXT	6	BOF BL	NA	NA	NA	NA	No Rep	NO
22PNINT	6	THR CON	NO	OK	OK	OK	No Rep	SI
22PNINTNIEVE	6	THR CON SNOW	NA	NA	NA	NA	No Rep	NO
22PNEXT	6	BOF CON	NO	OK	OK	NO	No Rep	SI
22PNEXTNIEVE	6	BOF CON SNOW	NA	NA	NA	NA	No Rep	NO

Validation Table by Sample (continued)

SampleID	month	Type	Ions balance Org	Conductivity	OrgN	Ratio Na/Cl	REP	VAL
25PHINT	6	THR CON	NA	NA	NA	NA	No Rep	NO
25PHEXT	6	BOF CON	NA	NA	NA	NA	No Rep	NO
26QIINT	6	THR BL	NO	NO	OK	OK	No Rep	NO
26QIEXT	6	BOF BL	NO	OK	OK	OK	No Rep	SI
30PSINT	6	THR CON	NO	OK	OK	OK	No Rep	SI
30PSINTNIEVE	6	THR CON SNOW	NA	NA	NA	NA	No Rep	NO
30PSINTLIX20	6	SW CON	NO	NO	OK	OK	No Rep	NO
30PSINTLIX60	6	SW CON	NO	NO	NA	NO	No Rep	NO
30PSEXT	6	BOF CON	NO	OK	OK	NO	No Rep	SI
30PSEXTNIEVE	6	BOF CON SNOW	NA	NA	NA	NA	No Rep	NO
33QPEINT	6	THR BL	NO	OK	OK	OK	No Rep	SI
33QPEINTNIEVE	6	THR BL SNOW	NA	NA	NA	NA	No Rep	NO
33QPEINTLIX20	6	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEINTLIX60	6	SW BL	NA	NA	NA	NA	No Rep	NO
33QPEEXT	6	BOF BL	NO	OK	OK	OK	No Rep	SI
33QPEEXTNIEVE	6	BOF BL SNOW	NA	NA	NA	NA	No Rep	NO
37PPRINT	6	THR CON	NO	OK	OK	OK	No Rep	SI
37PPREXT	6	BOF CON	NO	OK	OK	OK	No Rep	SI
54PHINT	6	THR CON	NO	OK	NO TN	OK	No Rep	NO
54PHEXT	6	BOF CON	NO	OK	NO TN	OK	No Rep	NO
102PPRINT	6	THR CON	NO	OK	OK	OK	No Rep	SI
102PPRINTLIX20	6	SW CON	NA	NA	NA	NA	No Rep	NO
102PPRINTLIX60	6	SW CON	NA	NA	NA	NA	No Rep	NO
102PPREXT	6	BOF CON	OK	OK	OK	OK	No Rep	SI
115FSINT	6	THR BL	NO	OK	OK	OK	No Rep	SI
115FSINTNIEVE	6	THR BL SNOW	NA	NA	NA	NA	No Rep	NO
115FSINTLIX20	6	SW BL	OK	OK	NA	OK	No Rep	NO
115FSINTLIX60	6	SW BL	OK	OK	NA	OK	No Rep	NO
115FSEXT	6	BOF BL	OK	OK	OK	OK	No Rep	SI
115FSEXTNIEVE	6	BOF BL SNOW	NA	NA	NA	NA	No Rep	NO